MIL-H-13109A
27 MAY 1966
SUPERSEDING
MIL-H-13109(SigC)
6 November 1953

MILITARY SPECIFICATION

HEADSETS, ELECTRICAL; GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for headsets used in military equipment. Specific requirements for a particular type of headset are listed in the applicable specification sheets (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder: Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76 - Tape: Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation, Methods Of.
MIL-T-152 - Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture and-Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).

FSC 5965

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129 - Marking For Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-169 - Extreme-Temperature Cycle.
MIL-STD-170 - Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-831 - Test Reports, Preparation of.

DRAWINGS

SIGNAL CORPS

SC-A-942 - Terminal TM-163.
SC-D-7723 - Headset HS-30-U Assembly.
SC-D-7724 - Receiver R-30-U Assembly.
SC-D-7725 - Receiver R-30-U Details.
SC-D-7726 - Receiver R-30-U Details.
SC-D-7727 - Receiver R-30-U Receiver Unit, Outline Dimensions.
SC-D-7728 - Cord CD-620-U Assembly.
SC-A-7729 - Cord CD-620-U Details.
SC-A-7730 - Cord Stay and Bushing, Details.
SC-D-7731 - Cord CD-620-U, Clip Assembly and Details.
SC-A-7732 - Solderless Terminal.
SC-D-7733 - 2 CC Coupler For W.E. Co. 640-AA Microphone.
SC-D-7976 - Headband HB-30, Assembly and Details.
SC-A-7978 - Insert M-300.
SC-D-7996 - Transformer C-410 Assembly and Details.
SC-A-7999 - Cord CD-605.
SC-B-8000 - Cordage for Cord CD-605.
SC-A-14702 - Sleeve, Telephone Cord.
SC-D-19953 - Cord CD-656-U Assembly.
SC-D-19954 - Cord CD-656-U Detail.
SC-D-19955 - Headset HS-29-U Assembly.
SC-A-21285-2 - Identification Tag Aluminum.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets and drawings. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

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3.2 Preproduction samples.

3.2.1 Quantities. The contractor shall furnish preproduction samples for approval if required by the invitation for bids or request for proposal and the contract (see 4.3 and 6.2).

3.2.2.1 Fabrication and assembly. Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect.

3.2.2.2 Approval. Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor, preproduction samples to be offered as units under the contract shall be refabricated, if necessary, by the contractor to meet specified requirements.

3.2.2.3 Deviation. When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, the reason therefore, and the manner in which each will be corrected in the production of the equipment under the contract.

3.2.3 Test data. The preproduction samples shall be accompanied by test data showing compliance with specified preformance criteria. The test data shall comprise an engineering report in accordance with MIL-STD-831, giving test procedures; observations; and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.2.4 Reference standards. After the preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in the custody of the government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding the interpretation of the requirements.

3.3 Construction. Unless otherwise specified herein, all construction shall be in accordance with the applicable specification sheet and the drawings listed in section 2.

3.4 Receiver R-30-U.

3.4.1 Receiver unit. The receiver unit shall consist of a diaphragm, permanent magnet and associated pole pieces, and coils or cords. The magnet pole pieces and coils shall be housed in a brass container. The back of the container shall be completely sealed.

3.4.1.1 Diaphragm. The material used in the construction of the diaphragm shall be Permendur or equivalent.

3.4.1.2 Permanent magnet. The permanent magnet shall be of precipitation-hardening alloy (alnico) or approved equivalent: cast or sintered, heat treated, and aged in accordance with the best commercial practice. It shall be capable of meeting the magnetic stability test of 4.6.4.

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3.4.1.3 Coil. The coil of the receiver shall be wound with class HF (heavy) Formex wire, as made by the General Electric Company, or equivalent. The area of the wire shall be not smaller than 9.9 circular mils (No. 40 AWG). If coils are used in the construction of the receiver unit, the wire shall be continuous (without splice) from the start of the first coil to the end of the second coil.

3.4.1.4 Cap and housing. The cap and housing of Receiver R-30-U shall be black molded plastic, type CFG, in accordance with MIL-M-14.

3.4.2 Impedance. The impedance of each receiver shall be within the limits of 105 to 145 ohms (nominal 128 ohms) when measured in accordance with 4.6.3.

3.4.3 Acoustical output. The receiver acoustical output shall be capable of passing the test of 4.6.2.

3.4.4 Dielectric strength and insulation resistance. The receiver unit shall be capable of withstanding 350 volts rms applied for 1 minute between the receiver terminal and the core. The insulation resistance measured with 500 volts dc applied shall be not less than 1 megohm. (See 4.6.1.)

3.5 Transformer C-410. Design and construction shall be in accordance with Drawing SC-D-7996 and the following requirements.

3.5.1 Impedance ratio. The impedance ratio between the winding shall be 1:31.2. (See 4.7.1.)

3.5.2 Insertion loss. The insertion loss of the transformer at 1,000 cycles per second shall not exceed 1.5 decibels. (See 4.7.2.)

3.5.3 The insertion loss at any frequency within the range of 400 to 3,500 cycles per second shall not differ more than 1.5 decibels from the insertion loss at 1,000 cycles per second.

3.5.4 Dielectric strength and insulation resistance. The transformer shall withstand a 60-cycle alternating current of 350 volts rms between windings for a period of 1 minute without showing evidence of breakdown. The insulation resistance between windings shall be not less than 1 megohm. (See 4.7.3.)

3.5.5 Impregnating and potting. The coils of the transformers shall be completely dried and thoroughly preheated and vacuum impregnated. The potting compound shall be F-835-S as made by Zophar Mills, Inc., Brooklyn, N.Y., or equivalent. Among the criteria of proper impregnation shall be uniform hardness, complete penetration of impregnant, and freedom from voids.

3.5.6 Soldering.

3.5.6.1 Solder. Solder used for electrical connections shall be composition Sn60 conforming to QQ-S-571.

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3.5.6.2 Flux and cleaning agents. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin; rosin and alcohol; or rosin and turpentine.

3.5.6.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. The soldering shall be so accomplished that no rosin remains on the joint or in the proximity of the joint. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment, flux being spread over a larger area, or resulting damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.6 Cleaning. Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning also shall be performed before final assembly of components.

3.7 Tropicalization. The assembled equipment shall be treated in accordance with MIL-T-152. Before being assembled into the equipment, materials and parts shall be treated in accordance with MIL-V-173.

3.8 Service conditions.

3.8.1 Equipment operating. The headsets shall meet the requirements of this specification under any probable combination of the following:

(a) Temperature - Any ambient temperature in the range of -40° F to 131° F.
(b) Relative humidity - Any relative humidity up to 98 percent.
(c) Sand and dust - Sand and dust as encountered during heavy traffic over dry sandy or clay roads. (See 4.9.4.)
(d) Drop - A drop of 6 feet on a concrete surface. (See 4.9.5.)

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3.8.2 Equipment nonoperating. The headsets shall meet the requirements of 3.8.1 after subjection to any one or more of the following nonoperating conditions:

(a) Temperature - Continuous exposure for 72 hours at -80° F and for 72 hours at +160° F, at any orientation.
(b) Relative humidity - Relative humidity up to 100 percent, including condensation caused by temperature changes.
(c) Immersion - The headset shall show no evidence of water in any cavity, including the transformer.

3.9 Marking.

3.9.1 Identification of product. Equipments, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9.2 Serial numbers. Serial numbers are not required for equipments covered by this specification.

3.10 Workmanship. The headsets shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.1.1.1 Inspection, general. Government inspection shall consist of group A and B inspection as specified in 4.4.1.1 through 4.4.3 (see 6.3.1). Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B inspection.

4.2 Classification of inspection. The examination and testing of the headsets shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
    1. Inspection of product for delivery (see 4.4.1).
    2. Inspection of preparation for delivery (see 4.4.3).

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4.3 Preproduction inspection. Preproduction inspection shall be performed by the supplier after award of contract and prior to production, at a location acceptable to the Government (see 6.2).

4.3.1 Sample and test routine. This inspection shall consist of the preproduction inspection specified in table I, groups I and II. Twelve samples will be subjected to the preproduction inspection for group I. Upon successful completion of group I inspection, the completed headsets shall be divided into 6 equal groups. Each group shall contain 2 samples and be subjected to the preproduction inspection for group II.

TABLE I. Preproduction inspection

Examination or test	Requirement paragraph	Method paragraph
Group I (12 sample units)		
Visual and mechanical (receiver units, transformer, and completed headsets)		4.5.1
Receiver R-30-U:		
Dielectric strength and insulation resistance	3.4.4	4.6.1
Impedance	3.4.2	4.6.3
Acoustical output	3.4.3	4.6.2
Transformer C-410:		
Dielectric strength and insulation resistance	3.5.4	4.7.3
Impedance ratio	3.5.1	4.7.1
Insertion loss	3.5.2	4.7.2
Completed headsets:		
Listening test		4.8
Group II (2 sample units)		
Completed headsets:		
Temperature	3.8.1 and 3.8.2	4.9.1
* Moisture resistance	3.8.1 and 3.8.2	4.9.2
* Immersion	3.8.2	4.9.3
* Sand and dust	3.8.1	4.9.4
* Drop	3.8.1	4.9.5
Receiver R-30-U:		
Magnetic stability	3.4.1.2	4.6.4

* When preproduction inspection of individual R-30-U earphone elements is required, the cord opening shall be blocked during the sand and dust, immersion, and moisture resistance tests. The drop test shall be performed with the earphone element mounted in an appropriate headband.

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4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and B inspections.

4.4.1.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. The record shall be presented to the Government when the items are resubmitted and shall become the property of the Government. Rejected lots shall be inspected using tightened inspection.

4.4.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II and MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the government inspector, except that the listening test shall be last.

4.4.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table II.

TABLE II. Group A inspection

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)
Visual and mechanical: (receiver units, transformer, and completed headsets) - - - - - - - - - -	---	4.5.1	1% major and 4% minor
* Electrical:			
Receiver R-30-U:			
Dielectric strength and insulation resistance - - - - -	3.4.4	4.6.1	1% for the group
Impedance - - - - - - - - - - - - - -	3.4.2	4.6.3	
Acoustical output - - - - - - - - - -	3.4.3	4.6.2	
Transformer C-410:			
Dielectric strength and insulation resistance - - - - -	3.5.4	4.7.3	1% for the group
Impedance ratio - - - - - - - - - -	3.5.1	4.7.1	
Insertion loss - - - - - - - - - - -	3.5.2	4.7.2	
Listening test (completed headset) - - -	---	4.8	1%

* All electrical defects shall be considered major.

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4.4.2 Group B inspection. Group B inspection shall consist of the examination and test specified in table III and shall be performed on sample units that have been subjected to and met group A inspection.

4.4.2.1 Sampling plan. One earphone element for group B inspection shall be selected, without regard to quality, from the first production lot submitted for inspection, and one sample for each 250 units, or fraction thereof, produced thereafter.

TABLE III. Group B inspection

Examination or test	Requirement paragraph	Method paragraph
Receiver R-30-U:		
Magnetic stability - - -	3.4.1.2	4.6.4

4.4.2.2 Reinspection of conforming group B sample units. Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract, provided they are resubjected to and pass group A inspection after repair of all visible damage.

4.4.2.3 Noncompliance. If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until correction action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the contracting officer.

4.4.3 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with MIL-P-116, table III, including rough handling tests, and the other specifications referenced herein, to verify conformance with the requirements in section 5 of this specification.

4.5 Test methods.

4.5.1 Visual and mechanical inspection. Parts and equipment shall be examined for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be examined for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc., shall be examined for workmanship (see 3.10). Examination also may be made for other visual and mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105.

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4.6 Receiver R-30-U.

4.6.1 Dielectric test and insulation resistance. A dielectric test shall be made on the receiver unit removed from the housing. The insulation between each receiver terminal and metal container or magnet core shall withstand the application of 350 volts rms ac at 60 cycles for 1 minute without breakdown. The insulation resistance of the receiver shall be measured with an applied voltage of 500 volts dc and shall be not less than 1 megohm. (See 3.4.4.)

4.6.2 Acoustical output. The acoustical output of the receivers shall be measured at 2 levels of electrical input power: 6 microwatts and 15.6 milliwatts. When the series combination consisting of the receiver and a 128-ohm noninductive resistance is supplied with a voltage of .0554 volts rms, at each test frequency, the input electrical power to the receiver is assumed to be 6 microwatts. For the test at the 15.6 milliwatt power level the voltage supplied to the series combination shall be maintained at 2.82 volts rms for each test frequency. For these tests the 2 cc coupler shall be located on top of the Western Electric Co. 640-AA Microphone, or equivalent, and to insure a good seal, the receiver unit shall be held in place on top of the 2 cc coupler by a 200-gram nonmagnetic weight placed on top of the receiver housing. The microphone shall be calibrated by the reciprocity method. (See 3.4.3.)

4.6.2.1 Power-input level: 6 microwatts. The receiver shall be connected in series with a noninductive resistance of 128 ohms. A voltage of .0554 volt rms, at each test frequency, shall be maintained across the series combination. The acoustical output of the receiver, measured in decibels relative to 0.0002 dyne per square centimeter, shall meet the requirements of table IV, 4.6.2.2, and 4.6.2.3.

TABLE IV. Response requirements at 6-microwatts power-input level

Frequency (cycles per second)	RMS pressure (in db relative to 0.0002 dyne per sq cm)	
	Column A	Column B
400	87.5	89.5
600	88.5	90.5
800	89.5	91.5
1,000	92	94
1,200	93	95
1,400	93	95
1,600	91.5	93.5
1,800	90	92
2,000	89	91
2,200	89	91
2,400	88	91
2,600	85	88
2,800	80.5	83.5
3,000	76.5	79.5
3,200	73	76
3,400	70.5	73.5
3,500	68.5	71.5

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4.6.2.2 Allowable percent of defective units of 6-microwatt power input level. Not more than 20 percent of the sample receivers shall have an output response less than specified in column B of table IV. No receiver shall have an output response less than specified in column A of table IV.

4.6.2.3 Frequency variation limits. The frequency shall be continuously varied from 400 to 3,500 cycles per second. From 400 to 2,200 cycles per second, the maximum allowable difference in response between any 2 frequencies in this range shall not exceed 8 decibels, and the maximum allowable difference in response between any 2 frequencies not more than 100 cycles per second apart shall not exceed 2.5 decibels. From 2,200 to 3,500 cycles per second, the maximum allowable difference in response between any 2 frequencies in this range shall not exceed 30 decibels; and the maximum allowable difference in response between any 2 frequencies not more than 100 cycles per second apart, shall not exceed 3.5 decibels.

4.6.2.4 Power-input level: 15.6 milliwatts. The receiver shall be connected in series with a noninductive resistance of 128 ohms. A voltage of 2.82 volts rms at each test frequency shall be maintained across the series combination. The acoustical output of the receiver measured in dynes per square centimeter shall be such as to meet the requirements of table V.

TABLE V. Response requirements at 15.6-milliwatts power-input level

Frequency (cycles per second)	Minimum RMS pressure (in db relative to 0.0002 dynes per sq cm)
400	122
600	124
800	126
1,000	126
1,200	126
1,400	126
1,600	125
1,800	124
2,000	123
2,200	122
2,400	121
2,600	118
2,800	113
3,000	109
3,200	106
3,400	101
3,500	100

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4.6.3 Impedance test. The impedance shall be measured as follows: The receiver shall be connected to the coupler and microphone as specified in 4.6.2. The cord terminals shall be connected to a source of alternating voltage. A noninductive resistance (approximately 1,000 times larger than the rated impedance of the receiver) shall be inserted in series with the source. The voltage of the source shall be adjusted to provide 0.025 volt (as measured by an electronic voltmeter) across the receiver at 1,000 cycles per second. A variable noninductive resistance shall then be inserted in the circuit in place of the receiver. The resistance shall be varied to obtain an electronic voltmeter reading of 0.025 volt with the voltage output of the source remaining as previously adjusted. The value of the resistance necessary to obtain the 0.025 volt reading shall be considered to be the impedance of the receiver. (See 3.4.2.)

4.6.4 Test for magnetic stability. The test for magnetic stability shall consist of subjecting the receiver to 6 successive discharges of opposite polarity from a 1 microfarad capacitor charged to 90 volts. The receiver shall be tested for acoustical output after this test. The response shall not have varied more than 2.5 decibels when tested in accordance with 4.6.2.1. (See 3.4.1.2.)

4.7 Transformer C-410.

4.7.1 Impedance ratio. The transformer shall be tested to show compliance with 3.5.1 when tested in the following manner: A noninductive resistance of 3,000 ohms shall be connected in series with the oscillator. The low-impedance (256 ohms) winding of the transformer shall be connected across this series combination. The oscillator shall be adjusted to provide a voltage of 1.5 ac volts across the winding. The ac voltage across the high-impedance (8,000 ohm) winding of the transformer shall be measured. The impedance ratio shall be determined from the formula:

$$\frac{Z_{low}}{Z_{high}} = \frac{V^2_{low}}{V^2_{high}}$$

All voltage measurements shall be made by means of a vacuum-tube voltmeter of not less than 1/2-megohm input impedance.

4.7.2 Insertion loss. The transformer shall be tested at frequencies of 400, 1,000, 1,800, and 3,500 cycles per second by the following method (see 3.5.2).

4.7.2.1 Test method. The impedance of the power source shall be 256 ohms. A noninductive resistance of 256 ohms shall be connected to the power source and the power level shall be adjusted across the 256-ohm noninductive resistance to 31.2 milliwatts. With a noninductive resistance of 8,000 ohms connected across the high-impedance (8,000 ohm) winding of the transformer under test, the power level of 31.2 milliwatts shall be transferred to the low-impedance (256 ohm) winding of the transformer. The voltage across the 8,000-ohm noninductive resistance shall then be measured. The following equation shall be used:

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The insertion loss in decibels equals 10 times the logarithm of the ratio of 31.2 milliwatts to the number of milliwatts supplied to the 8,000-ohm resistor. All voltage measurements shall be made by means of a vacuum-tube voltmeter of not less than 1/2-megohm input impedance. The transformer shall meet the requirements of 3.5.2.

4.7.3 Dielectric strength and insulation resistance test. A 60-cycle alternating current of 350 volts rms shall be applied to one end of the primary winding and to one end of the secondary winding for a period of one minute. There shall be no breakdown. The insulation resistance shall be measured using a 500-volt direct current applied continuously for 1 minute between windings. The insulation resistance shall be not less than 1 megohm. (See 3.5.4.)

4.8 Listening test. The headset shall be subjected to a listening test at the 6-microwatt input level and shall receive an intelligible signal.

4.9 Service condition tests.

4.9.1 Temperature. The headset shall be subjected to the temperature cycle shown in MIL-STD-169, except that the headsets shall be tested for electrical requirements at the conclusion of the temperature-cycling test. The acoustical output at the conclusion of the temperature-cycling test shall be not less than the minimum requirements of 4.6.2.1.

4.9.2 Moisture resistance. The completely assembled headset shall subjected to five 48-hour cycles of the standard humidity test as shown in MIL-STD-170. At the conclusion of this test, all components shall be disassembled and examined. There shall be no evidence of warping or other damage. Corrosion, if present, shall not be sufficient to interfere with proper operation of the equipment. The acoustical output at the conclusion of the humidity test shall be not less than the minimum requirements of 4.6.2.1.

4.9.3 Immersion. The headset shall be submerged in water to a depth of 3 feet for 2 hours. Upon removal from the water, the cap shall be unscrewed and the moisture wiped from the diaphragm and the receiver cap. The receiver shall then meet the test of 4.8.

4.9.3.1 The cap shall be removed from the receiver and the acoustic-resistance screen shall be allowed to dry for 20 minutes. After screwing the cap back in place, the acoustical output of the receiver unit shall be not less than the minimum requirements of 4.6.2.1.

4.9.4 Sand and dust test. The headset shall be tested as follows:

(a) Place the headset in its normal operating position, in the test chamber. Raise and maintain the sand and dust density of 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

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b. The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, U. S. standard sieve series	Percent
100 - mesh	100
140 - mesh	98 ±2
200 - mesh	90 ±2
325 - mech	75 ±2

c. Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition loss factors contained in above	0 to 2

d. Maintain the internal temperature of the test chamber at 77 ±4° F for a period of 6 hours, with the velocity of the sand and dust through the test chamber maintained between 100 and 500 feet per minute.

e. After 6 hours at above conditions, raise the temperature to 160 ±4° F and maintain it at this value for 6 hours. The other conditions shall remain the same.

f. Remove the headset from the test chamber and allow it to cool at room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the headset. However, dust shall not be removed by either blast or vacuum cleaning. The headset shall meet the minimum requirements of 4.6.2.1.

4.9.5 Drop test. The headset shall be dropped 6 times at random from a height of 6 feet. Impact at bottom shall be made on concrete. The headset shall receive intelligently after the test. There shall be no mechanical failure such as cracking or breaking of parts. Minor chipping shall not constitute a failure. The headset shall not suffer degradation of more than 2 decibels in its performance. (See 3.8.1.)

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Headsets shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Headsets shall be dried in accordance with one or more of the procedures listed in MIL-P-116. The procedure used shall not be injurious to the item.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Unless otherwise specified, headsets shall be adequately cushioned and individually packaged in accordance with MIL-P-116, method IA8, in a manner to comply with 3.5 and 3.6 therein. Headsets, so packaged, shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Headsets shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This package may conform to the commercial practice of the supplier for retail distribution, when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The packaged item(s) shall be packed in fiberboard containers conforming to PPP-B-636, weather resistant, style optional, special requirement. In lieu of the closure and waterproof requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, ends, edges, and manufacturer's joints with waterproof tape, 2 inches wide, conforming to PPP-T-60, class 1; or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636, using nonmetallic or tape banding only.

5.2.2 Level B. The packaged item(s) shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style and use requirements optional. Closure shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged item(s) shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This pack shall conform to the rules and regulations of the applicable carrier and may be the commercial practice of the supplier when such conforms to the requirements of this level.

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5.3 Marking (see 6.1). In addition to any special marking required by the contract or order, each unit package, intermediate, and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required, and shall contain equal quantities of identical items to the greatest extent possible.

6. NOTES

6.1 Intended use. These headsets are used in noisy locations as receivers for radio and wire equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of specification. (See 3.1.)
(b) Type required.
(c) Levels of preservation, packaging, and packing; and applicable marking (see sec. 5).
(d) Number of preproduction samples required (complete headsets - 12 each).
(e) Marking and shipping of samples.
(f) Place of preproduction inspection. (See 4.3.)
(g) Place of final inspection.

6.3 Contractual requirements. It is recommended that the following be included in the contract:

6.3.1 Additional testing. The Government reserves the right to test additional preproduction samples when considered necessary by the government inspector. It should be understood that additional inspection and nondestructive testing of the equipment may be performed by the government inspector when considered necessary to determine compliance with this specification and other applicable documents. The government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or at the plant of the contractor.

6.3.2 Waiver of preproduction samples. When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that: (a) the specified samples have been submitted and approved on the current contract; (b) production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and (c) current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent upon renegotiation of the contract to reflect the saving in cost to the contractor.

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6.4 Indirect shipments. The preservation, packaging, packing, and marking requirements specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - None
Air Force - 85

Review activities:
Army - EL
Navy - None
Air Force - 85

User activities:
Army - None
Navy - MC, CG
Air Force - 17

Preparing activity:
Army - EL

(Project 5965-0068)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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HEADQUARTERS COMMAND
U. S. ARMY ELECTRONICS [redacted]
FORT MONMOUTH, NEW JERSEY 07703

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Electronics Command
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Fort Monmouth, New Jersey 07703

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